Memory Match: The Design Rules in Plain English

This guide explains the rules in `CLAUDE.md` for readers who don't write code. `CLAUDE.md` is the instruction sheet that Claude (the AI assistant) follows while building this project. Think of it as the **project brief** you would give to a builder: what to build, what materials are allowed, and what is off-limits.

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1. What we're building, and why

This is a **classroom demonstration**. The goal is to show students, live, how software goes from an idea to a working product.

The class chose to build **Memory Match**, the classic card game:

- A grid of cards is laid face down. Each picture appears on exactly two cards.
- You turn over two cards at a time. If they match, they stay face up. If not, they flip back after one second.
- You win when every pair has been found.

Who it's for: anyone, of any age, playing casually.

What the finished game includes:

Feature	What it means for the player
Flip & match	The core game described above
Moves and timer	Shows how many tries you've taken and how long you've played
Win message and restart	A "You won!" box, plus a button to shuffle a new game
Difficulty levels	Easy (12 cards), Medium (16 cards), Hard (24 cards)

The pictures on the cards are **animal emoji** (🐶 🐱 🦊 🐼 …). Emoji are built into every device, so we don't need to supply any image files.

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2. The three non-negotiable rules

These rules apply to the project whatever it turns out to be. If an idea breaks one of them, the idea changes, not the rule.

Rule 1: Everything lives in a single file

The whole game is **one file** (`index.html`). You open it by double-clicking it, like a document, and it opens in your web browser.

- **No installing anything.** There's no app store, no setup program, and no account.
- **No internet needed.** It works on a plane, or in a classroom with bad Wi-Fi.
- **Easy to share.** You can email it, put it on a USB stick, or post it to a class page.

> 🧰 **Analogy:** a self-contained board game in one box, with nothing sold separately.

The rules allow one exception. If a future project needs larger pictures or sounds, they may go in a single `assets` folder next to the file, and the two are shared together as one zip. Each of those files must be **ours, or free to share**, and its source must be written down in a credits list. Memory Match didn't need this: it really is just one file.

Rule 2: The game never talks to the internet

The game never sends or receives anything online.

- It doesn't load fonts, pictures, or code from other websites.
- It has no ads, no analytics, and no "phone home".
- It has no AI features calling out to online services.

Why it matters: - **Reliability.** A website that changes or goes down elsewhere can't break the game. - **Privacy.** If nothing leaves the computer, nothing can leak. - **Teaching.** Everything the game does is visible in the one file, so the class can read all of it.

> 🏝️ **Analogy:** the game is an island with no bridges. Nothing comes in and nothing goes out.

Rule 3: No personal information, ever

The game must never **ask for, store, or display** personal details. That includes:

- names, email addresses, phone numbers, or home addresses

- student ID numbers or grades (school records are protected by law; in the U.S. that law is called FERPA)
- health, money, or government-ID details
- passwords or logins of any kind
- access to the camera, microphone, location, or contacts

There is also no tracking and no cookies. Any example data must be **obviously made up** (e.g. "Player 1"), never a real student's or teacher's name, even as a placeholder.

The rules do allow the game to remember **harmless things**, like a preferred setting or a best score labeled "You", on the player's own computer. Memory Match currently remembers nothing at all: close the tab and it's a fresh start.

> 🔒 **Why so strict?** This project is shown in a classroom and shared with students. The safest personal data is the data we never collect.

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3. Built to be read aloud

Students will watch the project being built and read along, so `CLAUDE.md` asks for instructions that are easy to follow:

- **Clear over clever.** Names describe what things are (`moves`, `startTimer`), not cryptic abbreviations.
- **Labeled sections.** The file is organized like a textbook chapter: appearance first, then the page layout, then the game's behavior, each under its own heading.
- **Notes explain *why*.** Short notes in the file answer the questions a student might ask, like "why is this here?"
- **Fewer, finished features.** It's better to have four features that work well than ten half-done ones.

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4. Designed for a projector and for everyone

The game has to work on a classroom projector, on a phone, and for people who use their computers in different ways.

Requirement	Why	How Memory Match does it
Big text, strong contrast	It must be readable from the back of the room	Large fonts, dark text on a light background
Works on any screen size	Some students play on phones	The card grid shrinks to fit, with no sideways scrolling
Works without a mouse	Some people use only a keyboard	Tab moves between cards, Enter flips them, and a bright outline shows where you are
Works with screen readers	Blind and low-vision players use software that reads the screen aloud	Each card announces itself, e.g. "Card 3, face down"
Color is never the only clue	Some people are color-blind	Matched cards turn green **and** show a ✓ mark
Explains itself	A first-time player shouldn't need instructions	A one-line description sits under the title
Dark mode (optional)	Easier on the eyes in a dark room	It switches automatically when the device is set to dark mode

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5. How the project gets built

`CLAUDE.md` sets out five stages. This is the path the class followed:

1. **Discuss & decide.** The class chose the game, the audience, and the features. The decisions were written into `CLAUDE.md` *before* any building started.
2. **Plan.** The work was split into small steps, and each step produced something visible in the browser.
3. **Build step by step.** After every step the game had to open and work. Each step was explained as it happened.
4. **Review.** The finished file is checked against a checklist (see below).
5. **Share.** The file is handed out.

Saving checkpoints ("version control")

After each working step, the project is saved as a **checkpoint** with a short plain-English note, like *"Add move counter and timer"*. This works like save points in a video game:

- The class can look back and see how the game grew, one step at a time.
- If something breaks during a live demo, we can return to the last good checkpoint.
- Every checkpoint is a working game. We never save a broken one.

The checkpoints stay on the teacher's computer. Nothing is uploaded online.

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6. The pre-share checklist, in plain words

Before the game is handed out, someone confirms each of these:

- ☐ It's **one file**, plus at most one `assets` folder.
- ☐ It **works with Wi-Fi turned off**.
- ☐ The file contains **no web addresses** and **no code that reaches the internet**.
- ☐ Every extra file (if any) is **actually used** and **credited**.
- ☐ No extra file contains **hidden personal details**, such as the location data some photos carry.
- ☐ The game **never asks for personal information**.
- ☐ All sample names and data are **clearly made up**.
- ☐ The browser reports **no errors**.
- ☐ It's **readable on a projector** and **usable on a phone**.
- ☐ It can be **played with the keyboard alone**.

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7. When someone asks for something that breaks a rule

The rules also say what to do when an idea conflicts with them: **say so plainly, and offer an alternative that fits the rules.** For example:

Request	Problem	Alternative that follows the rules
"Show the live weather on the cards"	Needs the internet (Rule 2)	Use a built-in set of weather emoji ☀️🌧️❄️
"Let students log in to save their scores"	Logins are personal data (Rule 3)	Save one best score on this computer, labeled "You"
"Put our class photos on the cards"	Photos of real people (Rule 3)	Use drawings, emoji, or free-to-use images
"Add a leaderboard with everyone's names"	Names plus online sharing (Rules 2 and 3)	A local "Player 1 vs Player 2" mode on one screen

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This guide summarizes `CLAUDE.md`. If the two ever disagree, `CLAUDE.md` is the official version.